



Rocks — FULL MIND MAP

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Chapter 03_Rocks

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Igneous Rocks — Formation aur Types

- **Igneous rocks** = lava/magma ke cooling-solidification se banti hain. 2 types: **Extrusive** (surface pe cool hoti hain, e.g. Basalt) aur **Intrusive** (crust ke andar cool hoti hain, e.g. Granite). [Q1]
- Igneous rocks **crystalline aur non-crystalline** dono ho sakti hain; inme fossils/layers generally absent hote hain, aur silicate minerals se largely composed hoti hain. [Q6]
- Fossils milne ka chance sabse kam: **Granite** (igneous rock hai) — Conglomerate, Sandstone, Shale (sab sedimentary) mein fossils mil sakte hain. [Q7]

Sedimentary Rocks — Properties

- Sedimentary rocks: hydrological system se banti hain, pre-existing rocks ke weathering se involve hoti hain, **fossils contain karti hain**, aur **layers/strata** mein occur karti hain — sab 4 statements correct hain. [Q2, Q3]
- Sandstone layered hoti hai kyunki uske **sediments successive layers mein deposit + compact** hote hain (sirf desert/mountain mein nahi banti, rivers/beaches/seas mein bhi ban sakti hai). [Q4]

Porosity vs Permeability — Chalk vs Clay

- Chalk aur Clay dono **porous** hote hain, lekin Chalk **highly permeable** hai (cracks/fractures ki wajah se aquifer ban sakta hai) jabki Clay **impermeable/aquitard** hai (pores interconnected nahi hote). [Q5]



EXAM TRAP

- Clay 'not porous' claim GALAT hai — Clay actually highly porous hoti hai, lekin permeability extremely low hoti hai. Porosity aur Permeability alag concepts hain. [Q5]

Metamorphic Rocks — Origin

- **Metamorphic rocks** = 'change of form' — **Igneous AND Sedimentary** dono rocks se ban sakti hain, heat + pressure ke through. [Q8]
- Examples: Granite→Gneiss, Basalt→Schist/Amphibolite (igneous se); Shale→Slate, Limestone→Marble, Sandstone→Quartzite (sedimentary se). [Q9]

- Granite metamorphic rock **NAHI** hai — yeh igneous rock hai (na ki Marble/Quartzite/Slate jaisi metamorphic). *[Q9]*
- Odd-one-out: **Gneiss** (metamorphic hai) — Limestone, Sandstone, Shale sab sedimentary group ke hain. *[Q10]*

| Weathering — Rainfall ka Role

- Rainfall rocks ke weathering ka ek cause hai kyunki rainwater mein **dissolved CO₂ (carbonation) AND dissolved oxygen (oxidation)** dono hote hain — dono processes chemical weathering explain karte hain. *[Q11]*